Demonstrating Online Schema Alignment in Decentralized Knowledge Graph Querying

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Abstract. The integration of heterogeneous Knowledge Graphs remains challenging, particularly in decentralized environments where discovery queries executed over distributed, unindexed sources requiring alignment rules for unified querying.**RT**: This first sentence is very loaded. I recommend splitting it up into multiple sentences, with some more context on what every part means, possible with some examples. This demo paper presents an online schema alignment approach integrated into a Link Traversal Query Processing engine using a web-based interface. A social media scenario demonstrates querying across networks with varying vocabularies. Source code, library, and command-line interface are provided for integration into other applications. **RT**: In general, the abstract doesn't feel very exciting to me. Can you try formulating it using our standard context,need,task,... format?

Keywords: SPARQL · Decentralization · Link Traversal · Schema Alignment

1 Introduction

Integrating multiple Knowledge Graphs (KGs) presents scalability and heterogeneity challenges. Scalability constraints due to data size or privacy **RT**: Another reason is that KGs are simply often managed/created by different orgs/teams/companies, and federation allows you to cross these organizational boundaries. I'd mention this as well. often necessitate federated querying or link traversal-based querying [5] rather than materialization. Heterogeneity in knowledge representation and vocabulary poses another major challenge, as query issuers expect to formulate queries under a single, consistent representation. **RT**: The connection between these sentences is missing. Make sure to first mention the need for schema alignment/reason, and how it can solve the problem above. Schema alignment can be viewed as a subset of the problem of reasoning. Prior work has investigated reasoning over SPARQL endpoints [13], RDF streams [3], and schema alignment in SPARQL systems [4, 6, 8]. However, existing approaches assume alignment rules are known and scoped a priori. Our work introduces *online schema alignment*, where schema mappings are discovered, scoped, and applied dynamically during query execution of Link Traversal Query Processing (LTQP) [5].**RT**: I like this part, but the reader does not not LTQP yet. So

make sure to explain it (and why it's needed) briefly before you get to this part. In LTQP over unindexed distributed networks, different providers often express semantically equivalent properties heterogeneously. Even trivial inconsistencies (e.g., HTTP vs. HTTPS schema.org vocabularies) require query adaptations, like `UNION` clauses.**RT**: Let's explicitly use the term "query rewriting" here. However, this approach has critical limitations, as it assumes query issuers know all relevant alignment rules across large decentralized networks, and that rules apply globally, which is unrealistic given ontology design complexity and that mistakes can be made by data providers [2, 7]. Deeper semantic mismatches further complicate matters. For example, providers may conceptualize responses as "replies" or "comments", or represent attributes like gender or city names as literals or IRI. Rule chaining compounds the problem by requiring either materializing intermediate KGs through multiple `CONSTRUCT` queries or backchaining to materialize a final query [8]. Both methods are incompatible with LTQP. Executing multiple queries removes awareness of already-traversed documents, while backchaining into a single final query fails to discover intermediate resources needed for traversing the required search domain.

The rest of this paper presents the proposed approach **RT**: But what is the proposed approach? It has not been hinted to yet., the demonstrator, and conclusions.

2 Online Schema Alignment

Schema alignment requires the definition of a set of rules. In this work, we draw [1] on the Simple Standard for Sharing Ontological Mappings (SSSOM) specification [1, 9] to structure and formalize these rules.[2] An example of the resulting serialization is available online [3].

For the purpose of this work, we developed a simplified version of SSSOM and use the concept of *subweb* [12] to specify the domain of applicability of the schema alignment rules. In other words, these rules apply only within the designated set of IRIs. We opted for this design for several practical reasons.**RT**: Are the following paragraphs the two reasons? If so, start them with "First," and "Second,".

Our system operates under the open-world assumption, in which multiple new KGs and schema alignment rules may be discovered during processing. To ensure robustness, the system must prevent infinite recursive rule applications.[4] When such situations occur, it should ideally resolve them without aborting the entire computation. Since rules are applied within specific domains, the likelihood of infinite recursion is significantly reduced. In cases where infinite recursion is

[1] `https://onlineschemaalignmentltqp.github.io/vocabulary/vocab.ttl`

[2] We adopt this approach for simplicity, as the focus of this paper is not on the formatting of the rules but on their discovery and applications.

[3] `https://onlineschemaalignmentltqp.github.io/vocabulary/example.ttl`

[4] With our schema alignment entailment, logical contradictions are not possible.

detected or overlap of subwebs, the simple resolution strategy of rejecting the newly generated schema alignment rule is employed.

Another important consideration is the potential misuse, overspecification, and underspecification of ontology terms [2, 7]. In our use case, the goal is to enable querying data across the web under the assumption that data providers best understand their own data models and the intended entailments. Thus, we scope the rules to non-overlapping subwebs to prevent unintended entailments. However, we also allow query issuers to define additional rules that may interact with those provided by data publishers. This is justified by the fact that query issuers can more easily track and manage the consequences of their own entailment definitions, whereas it would be unrealistic to expect data providers to anticipate the broader implications of their rules beyond the scope of their own datasets or use cases.

To discover alignment rules, we modify the traversal policy **RT**: Traversal policies have not been explained yet. Either do this here, or in the intro. to follow object terms of triples with predicate IRI `semmap:ruleSetLocation`, where the dereferenced term provides a rule set. When the engine encounters a rule set at step s_i, it applies the rules to its internal KG G_{s_i} via forward chaining, producing an aligned graph G'_{s_i}. The triples of G'_{s_i}, rather than G_{s_i}, are passed to the traversal policy and inserted into the internal knowledge base for join processing, as traversal policies are not designed to handle vocabulary mismatches.**RT**: I would just explain this in words. The semi-formal notation adds no value IMO.

3 Demonstration

We implemented an online schema alignment system using the link traversal version of the Comunica [11] query engine. Both our demonstration [5] and implementation [6] are open source and publicly available. A video of the demonstration is available online. [7] Users can download the implementation to integrate it as a library or use the command-line interface to execute queries. Our demonstration scenario is based on a social media network built using the SolidBench benchmark [10]. This social media network is decentralized, meaning that users store their data in personal data vaults called *pods*, there is no centralized endpoint to query user information. For the purpose of this demonstration, we modified the user data such that different users express similar information using different vocabularies. Additionally, each user exposes a file that describes their schema alignment rules. The demonstrator is a web-based application that allows users to execute either the proposed queries or arbitrary custom queries. Queries can be run over two network configurations: the base network, where all pods use the same vocabulary, and the modified network, where pods use alternative vocabularies. The proposed queries are typical social media-related queries, such

[5] `https://github.com/onlineSchemaAlignmentLTQP/demo`

[6] `https://www.npmjs.com/package/query-sparql-link-traversal-solid-schema-alignment`

[7] `https://www.youtube.com/watch?v=fGDQwu65los`

as retrieving information about a user, identifying the forums where a user has posted, and finding posts liked by specific users. To illustrate, consider a query retrieving a user's profile, when executed over the modified network without alignment, the query returns incomplete results because some pods use HTTPS variants or alternative terms for the same properties. With online schema alignment enabled, the engine discovers the relevant mapping rules during traversal and returns the complete result set. For each execution, the system provides detailed feedback, including the query results, execution time, and the alignment rules discovered during query processing, along with their associated subwebs. For the queries executable under both configurations, the average overhead introduced by online schema alignment is 0.28 ± 0.31 seconds. **RT**: Can you also mention the appromiate average exec time of queries, so readers see that this overhead is not that significant. [8] Users can also define custom schema alignment rules and specify the subwebs to which these rules apply.

4 Conclusion

In this demo paper, we present a system for online schema alignment that employs a vocabulary inspired by the SSSOM specification [1, 9]. Through this demonstration, we show that, in unindexed and decentralized networks, LTQP queries can be executed successfully in a generalized manner without increasing the complexity of the user-issued queries. Future work includes extending the system to support more advanced reasoning mechanisms, such as the SPARQL entailment regimes and Notation3. **RT**: Maybe you can also hint here towards the WIP SHACL Rules spec Engines such as the EYE reasoner [14] or modern Prolog implementations like Scryer Prolog could enable such reasoning capabilities.

RT: What is missing for the in this conclusion, is a clear view on the limitations of the current work, and an idea on how these can be solved. **RT**: Also, make sure to end with some high-level perspectives. Come back to the need of this work. What is the impact? What does it mean for KG tools and applications? What is the impact on the broader society?

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